

MACHINE LEARNING

CORE CONCEPTS HANDBOOK

Artificial Intelligence (AI)

AI is the broader field that enables machines to simulate human intelligence.

Machine Learning (ML)

Machine Learning is a subset of AI that enables systems to learn patterns from data without being explicitly programmed.

Instead of writing rules → We provide data → Model learns patterns.

Types of Machine Learning

Supervised Learning

Learning from labeled data (X, y).
Used for: Classification & Regression.

Unsupervised Learning

Learning from unlabeled data.
Used for: Clustering & Association.

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Dataset Terminology

Dataset

Collection of data samples.

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Training Dataset

Data used to train the model.

Testing Dataset

Data used to evaluate the model.

Train-Test Split

Splits dataset into training and testing parts (commonly 80:20 or 70:30).

```
train_test_split(X, y, test_size=0.2)
```

Variables in Machine Learning

Feature (Independent Variable)

Input variable used to predict output.
Example: Weight, Surface.

Label (Dependent Variable / Target)

Output variable predicted by model.
Example: Tennis, Cricket.

-  Inbox

1,168
-  Starred
-  Snoozed
-  Sent
-  Drafts

7
-  Purchases

22
-  More
- Labels

+

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-  Starred
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+



Types of Variables

Categorical Variable

Represents categories (non-numeric labels).
Example: Surface type (Rough/Smooth).

Continuous Variable

Numeric variable with infinite possible values within range.
Example: Weight (30–110 grams).

Discrete Variable

Countable numeric values.
Example: Number of balls.

Classification vs Regression

Classification

Predicts categorical output.
Example: Tennis or Cricket.

Regression

Predicts continuous numeric output.
Example: Predict weight in grams.



Binary vs Multiclass Classification

Binary Classification

Two output classes.
Example: Tennis vs Cricket.

Multiclass Classification

More than two classes.
Example: Tennis, Cricket, Football.

Ball Classification Setup

Encoding:

Tennis = 1 (Positive Class)
Cricket = 2 (Negative Class)
Rough = 1
Smooth = 0

Features:

- Weight
- Surface

Confusion Matrix Structure

		Predicted	
		Tennis	Cricket
Actual	Tennis	TP	FN
	Cricket	FP	TN

True Positive (TP)

Correctly predicted positive samples.

True Negative (TN)

Correctly predicted negative samples.

False Positive (FP) – Type 1 Error

Incorrectly predicted positive when actual is negative.

False Negative (FN) – Type 2 Error

Incorrectly predicted negative when actual is positive.

Confusion Matrix

Suppose test results:

Ball	Actual	Predicted
1	Tennis	Tennis
2	Tennis	Cricket
3	Tennis	Tennis
4	Cricket	Cricket
5	Cricket	Tennis
6	Cricket	Cricket

More

Labels

+

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22

More

Labels

+

Tennis = Positive

True Positive (TP)

Actual = Tennis
Predicted = Tennis

Balls 1, 3
TP = 2

False Negative (FN)

Actual = Tennis
Predicted = Cricket

Ball 2
FN = 1

True Negative (TN)

Actual = Cricket
Predicted = Cricket

Balls 4, 6
TN = 2

False Positive (FP)

Actual = Cricket
Predicted = Tennis

Ball 5
FP = 1

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आम्ही Technical संस्कार करते !!!

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Type 1 & Type 2 Errors

Error Type	Meaning	Ball Example
Type 1 Error	False Positive	Cricket predicted as Tennis
Type 2 Error	False Negative	Tennis predicted as Cricket

Evaluation Metrics

Accuracy

Measures overall correctness.

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

Range: 0 to 1 (or 0%-100%)

Higher is better.

Precision

Measures correctness of positive predictions.

$$\text{Precision} = TP / (TP + FP)$$

Range: 0 to 1

High precision = Low false positives.

Recall (Sensitivity)

Measures ability to detect actual positives.

$$\text{Recall (Sensitivity)} = TP / (TP + FN)$$

Range: 0 to 1

High recall = Low false negatives.

Specificity

Measures ability to detect actual negatives.

$$\text{Specificity} = TN / (TN + FP)$$

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F1 Score

Harmonic mean of precision and recall.

$$F1 \text{ Score} = 2 * (\text{Precision} * \text{Recall}) / (\text{Precision} + \text{Recall})$$

Range: 0 to 1

Useful when dataset is imbalanced.

